

Audrey Houghton (she/her)

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RESEARCH INTERESTS

Scientific software engineering, reproducible computational workflows, astronomical data processing, geospatial analysis, high-performance computing, causal inference, machine learning for scientific data analysis, and large-scale data pipelines. Contributed to more than 30 scientific publications spanning astronomy, neuroimaging, psychiatry, and scientific software.

EDUCATION

University of Montana - Missoula, MT
Bachelors of Arts in Physics

Aug 2013 - May 2017
umt.edu/physics-astronomy

RESEARCH EXPERIENCE

University of Minnesota – Department of Psychiatry & Behavioral Sciences

Dec 2025 - Present

Research Software Engineer III

Minneapolis, MN (Remote)

- Develop and maintain Python analysis pipelines for GIS-linked electronic health record data in psychiatric research.
- Build geospatial feature-engineering workflows in Python/Jupyter for county- and tract-level overlays, geocoding quality control, and FIPS recovery from external APIs.
- Build publication-grade clinical prediction workflows with LightGBM, repeated stratified cross-validation, calibration selection, and bootstrap/permutation robustness checks.
- Lead statistical result interpretation and methods reporting for manuscripts and grant submissions across interdisciplinary teams.

Masonic Institute of the Developing Brain

Jan 2021 - Apr 2024

Research Software Engineer II (promoted to III)

Minneapolis, MN (Remote)

- Developed Python wrappers leveraging AWS S3 storage and SLURM optimization to accelerate image processing up to 12×.
- Engineered a containerized infant MRI segmentation application achieving 600× faster processing and improved segmentation accuracy.
- Built ETL pipelines for petabyte-scale neuroimaging datasets, emphasizing reproducibility and quality control.
- Authored laboratory documentation and technical curriculum for neuroimaging software and HPC workflows.
- Co-designed and instructed a three-month MRI Data Processing Workshop covering Bash, Linux, SLURM, BIDS, containers, and MRI analysis workflows.
- Founded and led weekly Bash training sessions for research staff and students.
- Founded and organized the interdisciplinary "Brainy Brains" seminar series to promote neuroscience learning across research disciplines.
- Mentored two University of Minnesota UROP/URA students in MRI data processing and reproducible workflows using ABCD and HCP-YA datasets.
- Processed and maintained workflows for major neuroimaging cohorts including ABCD, BCP, HCP-Development, HCP-Young Adult, ABIDE I, and ABIDE II.

Rutgers University Astrophysics Department

May 2016 – Aug 2016

Research Assistant

New Brunswick, NJ

- Refactored legacy computational astrophysics software from IDL to Python, improving maintainability and accessibility for future research.
- Developed and optimized galactic collision simulations through parameter estimation and computational modeling.
- Performed analysis and validation of simulation outputs to evaluate model performance and scientific accuracy.
- Collaborated with faculty and researchers on computational astrophysics projects using scientific programming and numerical methods.
- Presented research findings through an oral presentation to faculty and researchers in the Rutgers Department of Physics and Astronomy and a poster presentation at the interdisciplinary Summer Research Symposium Featuring REU and RiSE Scholars.

The MINERVA Project

Jan 2015 – May 2017

Research Assistant

Missoula, MT

- Developed a Python-based photometric data reduction pipeline for detecting transiting exoplanets from time-domain FITS imaging data.
- Implemented CCD calibration workflows, including bias, dark, and flat-field correction, followed by aperture and differential photometry for high-precision light curve generation.
- Utilized Astrometry, Astropy, Photutils, and the scientific Python ecosystem to automate astronomical image processing, quality control, and photometric analysis.

- Supported exoplanet transit detection and photometric follow-up as part of the MINERVA exoplanet research program.
- Collaborated with faculty and student researchers to develop and improve scientific software supporting astronomical research.
- Presented pipeline development and research results at the 2016 Montana Space Grant Consortium Conference.
- Developed software supporting published astronomical research, including a refereed journal article and an American Astronomical Society conference abstract.

INDUSTRY EXPERIENCE

CoreWeave, Inc.

Apr 2024 - Sept 2024

Fleet Operations Engineer

Remote

- Provisioned and operated thousands of NVIDIA GPU nodes in Kubernetes using Ansible, AWX, Grafana, and Go, achieving 1-2 week onboarding and 6+ months average production uptime.
- Diagnosed and resolved hardware and software failures across thousands of GPU compute nodes supporting large-scale AI workloads.
- Partnered with 10+ teams—including networking, hardware, operations, and support—to streamline troubleshooting, accelerate incident resolution, and document SOPs and post-mortems, improving efficiency and reliability.
- Identified points in the node lifecycle where nodes were lost, implementing improvements that saved the company millions in revenue.

Calyxt

Oct 2019 - Aug 2020

Laboratory Assistant & Data Specialist

Roseville, MN

- Developed Python scripts for laboratory data tracking, quality control, and analysis reporting.

TEACHING & MENTORING

University of Minnesota Undergraduate Research Opportunities Program (UROP/URA)

2023

Undergraduate Research Mentor

Minneapolis, MN

- Mentored two undergraduate researchers in neuroimaging data processing.
- Trained students to process Human Connectome Project Young Adult (HCP-YA) and ABCD MRI datasets.
- Introduced reproducible workflows and quality-control procedures for large neuroimaging datasets.

Co-Instructor, MRI Data Processing Workshop

2022

Masonic Institute for the Developing Brain, University of Minnesota

Minneapolis, MN

- Co-designed and taught a three-month workshop introducing students and researchers to neuroimaging data processing and high-performance computing.
- Developed instructional materials covering Bash/Linux, SLURM, S3 storage, BIDS standards, containers, quality assurance, and reproducible MRI processing workflows.
- Held weekly office hours and provided technical mentorship to participants from diverse academic backgrounds.
- The workshop was later adapted into an official University of Minnesota credit-bearing course.

Mathematics Teacher

Jul 2017 - Jul 2018

Imagine Prep Surprise

Surprise, AZ

- Taught Algebra II, Precalculus, Calculus, and Cambridge AS/A-Level Mathematics to high school students.
- Developed lesson plans, assessments, and instructional materials.
- Contributed to improving Algebra proficiency from approximately 11% to 70% during my tenure.

Teaching Assistant, Observational Astronomy

Aug 2016 - Dec 2016

University of Montana

Missoula, MT

- Assisted instruction for the undergraduate Observational Astronomy course.
- Supervised observing sessions and telescope operations.
- Delivered a 45-minute lecture on Python-based astronomical data reduction.
- Assisted students with programming and data analysis assignments.

Mathematics Instructor

Jul 2019 - Oct 2019

Huntington Learning Center

Plymouth, MN

- Provided one-on-one mathematics tutoring for K-12 students.
- Adapted instruction to students with varying backgrounds and learning styles.

Mathematics Instructor

Jan 2019 - May 2019

Tutor Doctor

Loveland, CO

- Delivered individualized mathematics tutoring to middle and high school students.
- Prepared students for coursework and standardized examinations.

Planetarium Outreach Educator

Aug 2015 - May 2017

University of Montana Physics Department

Missoula, MT

- Presented public planetarium shows introducing astronomy and physics concepts.
- Communicated scientific concepts to audiences of all ages.
- Supported departmental outreach activities to promote STEM education.

Peer-Reviewed Journal Articles

- Mehta, K., et al. (2024). XCP-D: A robust pipeline for the post-processing of fMRI data. *Imaging Neuroscience*, 2, imag-2-00257.
- Luo, A. C., et al. (2024). Functional connectivity development along the sensorimotor-association axis enhances the cortical hierarchy. *Nature Communications*, 15(1), 3511.
- Keller, A. S., et al. (2023). Personalized functional brain network topography is associated with individual differences in youth cognition. *Nature Communications*, 14(1), 8411.
- Pines, A., et al. (2023). Development of top-down cortical propagations in youth. *Neuron*, 111(8), 1316-1330.
- Keller, A. S., et al. (2024). A general exposome factor explains individual differences in functional brain network topography and cognition in youth. *Developmental Cognitive Neuroscience*, 66, 101370.
- Covitz, S., et al. (2022). Curation of BIDS (CuBIDS): A workflow and software package for streamlining reproducible curation of large BIDS datasets. *NeuroImage*, 263, 119609.
- Wilson, M. L., et al. (2019). First radial velocity results from the miniature exoplanet radial velocity array (MINERVA). *Publications of the Astronomical Society of the Pacific*, 131(1005), 115001.
- Zhao, C., et al. (2024). A reproducible and generalizable software workflow for analysis of large-scale neuroimaging data collections using BIDS apps. *Imaging Neuroscience*, 2, imag-2-00074.
- Byington, N., et al. (2023). Polynuro risk scores capture widely distributed connectivity patterns of cognition. *Developmental Cognitive Neuroscience*, 60, 101231.
- Shafiei, G., et al. (2024). Generalizable links between borderline personality traits and functional connectivity. *Biological Psychiatry*, 96(6), 486-494.
- Holt-Gosselin, B., et al. (2024). Familial risk for depression moderates neural circuitry in healthy preadolescents to predict adolescent depression symptoms in the Adolescent Brain Cognitive Development (ABCD) study. *Developmental Cognitive Neuroscience*, 68, 101400.
- Feczko, E., et al. (2025). Baby Open Brains: An open-source dataset of infant brain segmentations. *Scientific Data*, 12(1), 1423.
- Giovinazzi, M. R., et al. (2020). The HD 217107 planetary system: Twenty years of radial velocity measurements. *Astronomische Nachrichten*, 341(9), 870-878.

Conference Proceedings & Abstracts

- Shang, Z., et al. (2022). Learning strategies for contrast-agnostic segmentation via SynthSeg for infant MRI data. In *Proceedings of the International Conference on Medical Imaging with Deep Learning* (pp. 1075-1084).
- Richie-Halford, A., et al. (2022). NiRV: The neuroimaging report viewer. In *Proceedings of the Organization for Human Brain Mapping*.
- Houghton, A., et al. (2017). Project MINERVA's follow-up on wide-field, small telescope photometry to identify exoplanets. *American Astronomical Society Meeting Abstracts*, 229, 146.08.
- Keller, A., et al. (2024). Understanding cognitive development by capturing complex, multidimensional, childhood environments and individual-specific patterns of functional brain network organization. *Biological Psychiatry*, S85.
- Keller, A., et al. (2023). Highlighting the role of multidimensional childhood environments in functional brain network organization and cognitive development. *Neuropsychopharmacology*, 48, 134-135.
- Bertolero, M., et al. (2022). Multivariate patterns of functional connectivity are linked to borderline-spectrum symptoms in young adulthood and youth. *Biological Psychiatry*, 91(9), S170-S171.

Preprints & Manuscripts

- Goncalves, M., et al. (2025). FMRIPrep Lifespan: Extending a robust pipeline for functional MRI preprocessing to developmental neuroimaging. *bioRxiv*.
- Feczko, E., et al. (2024). Baby Open Brains: An open-source repository of infant brain segmentations. *bioRxiv*.
- Coffman, C., et al. (2024). Heritability estimation of subcortical volumes in a multi-ethnic multi-site cohort study. *bioRxiv*.
- Shafiei, G., et al. (2023). Generalizable links between symptoms of borderline personality disorder and functional connectivity. *bioRxiv*.
- Randolph, A. C., et al. (2026). Mental health outcomes of foster and adopted individuals with adverse childhood experiences: A validation of known risks using EHR data. *medRxiv*, 2026.05.28.26354276.
- Caola, L., et al. (2026). Before birth, beyond childhood: Understanding the influence of prenatal substance exposure on psychiatric diagnoses. *medRxiv*, 2026.05.27.26354275.
- Mahmoudi, M., et al. (2026). Cortical thickness and curvature in autism and ADHD: A mega-analysis. *bioRxiv*.

Research Pipelines

GIS Analysis Pipeline

University of Minnesota Department of Psychiatry & Behavioral Sciences

2026
[Research Software]

- Developed a reproducible geospatial analysis pipeline integrating county- and tract-level social determinants of health with electronic health record data.
- Automated geocoding quality control, FIPS recovery, census tract matching, and feature engineering using Python and external GIS APIs.
- Supported publication-quality geospatial analyses and interactive visualizations for psychiatric epidemiology research.
- Integrated Census tract and county-level indices (e.g., SVI, ADI) with electronic health record data to support reproducible spatial epidemiology analyses.

Clinical Prediction Analysis Framework

University of Minnesota Department of Psychiatry & Behavioral Sciences

2026
[Research Software]

- Transformed an existing psychiatric prediction pipeline into a modular, reproducible research framework by expanding functionality, standardizing workflows, and improving usability in accordance with FAIR software principles.
- Implemented automated repeated stratified cross-validation, LightGBM modeling, probability calibration, bootstrap/permutation robustness checks, SHAP interpretation, and publication-ready statistical reporting.
- Supported multiple psychiatric prediction studies through reusable, publication-ready software workflows.

Causal Learning Pipeline

University of Minnesota Department of Psychiatry & Behavioral Sciences

2026
[Research Software]

- Built a reusable Python pipeline integrating multiple causal discovery algorithms, graph visualization, effect estimation, and reproducible statistical workflows for psychiatric EHR analyses.
- Standardized preprocessing, graph generation, and downstream statistical analyses to support consistent comparison across causal learning methods.

SLURM Wrappers

High-Performance Computing S3 and Disk SLURM-Compliant Wrappers

2021 - 2024
github.com/DCAN-Labs/SLURM_wrappers

- Bash/Python wrappers adaptable to any SLURM-based data pipeline.

CABINET

Container-Linking Wrapper

2022 - 2023
github.com/DCAN-Labs/CABINET

- Python-based universal container-linking wrapper employing FAIR principles.

BIBSNet

Infant MRI Brain Segmentation Application

2022 - 2023
github.com/DCAN-Labs/BIBSnet

- Containerized infant MRI application; achieved 600x faster processing and +53% accuracy.

Research Infrastructure

ABCD Study Collection 3165 Data Processing

NIH Data Archive (NDA)

2021 - 2024
collection3165.readthedocs.io

- Community dataset enabling access to petabytes of processed/unprocessed MRI data under BIDS standards.

DCAN Labs Informational Guide

Minnesota Supercomputing Institute Data Engineering Documentation

2022 - 2024
dcan-labs-informational-guide.readthedocs.io

- Comprehensive, continually-supported documentation of usage of the Minnesota Supercomputing Institute to process, analyze, and transfer MRI data.

CoreWeave Internal Documentation

CoreWeave, Inc.

2024
[Proprietary]

- Initiated and built company-wide documentation from the ground up, replacing scattered one-off documents with an organized, centralized Notion knowledge base covering terminology, SOPs, and team workflows.

Independent Projects

Waste Not

Sustainability Application

2018 - 2022
[Independent Project]

- Designed and developed a sustainability application using a nationwide, self-curated recycling database created through large-scale web scraping, data normalization, and categorization.

TECHNICAL SKILLS

Programming: Python, Bash, UNIX/Linux, Git/GitHub, LaTeX, Docker, Singularity, SQL

Infrastructure & HPC: AWS (S3), Ceph, SLURM, Kubernetes, batch processing, data validation, Globus, Grafana, Prometheus, HPC workflow design

Machine Learning & Statistics: logistic regression, predictive modeling, LightGBM, calibration (Platt/isotonic), repeated stratified cross-validation, bootstrap/permutation inference, FDR correction (Benjamini-Hochberg), SHAP interpretability, ROC-AUC/PR-AUC/Brier metrics, causal inference & discovery

Scientific Computing: neuroimaging/MRI segmentation, astronomical image processing (FITS, Astrometry, Astropy, Photutils), CCD calibration, differential photometry, time-domain imaging, deep learning inference workloads and compute planning, GIS analysis, QGIS visualization, geocoding/FIPS recovery

Data Engineering: ETL pipelines, EHR integration, census tract and county-level feature engineering, data harmonization, reproducible research workflows, workflow automation

Collaboration: Jira, Linear, Kanban, Agile/Scrum, SOP documentation, Slack, Microsoft Teams, Google Workspace, Box, Salesforce API automation, compliant web scraping

Additional Exposure: Matlab, R, IDL, JavaScript/Node.js, Go, Ansible/AWX, RedHat, ReadTheDocs, Google Apps Script, GCP, Terraform, Datadog, CI/CD (GitHub Actions, CircleCI)

LEADERSHIP & SERVICE

Founder & Organizer, Brainy Brains Seminar Series

Masonic Institute for the Developing Brain, University of Minnesota | 2022–2024

- Founded and led a weekly interdisciplinary seminar where researchers from diverse backgrounds learned and presented topics in neuroscience.
- Fostered cross-disciplinary collaboration and scientific communication across the laboratory.

Founder & Organizer, Weekly Bash Training Series

Masonic Institute for the Developing Brain, University of Minnesota | 2021–2024

- Founded and led weekly Bash and Linux training sessions for researchers and students.
- Developed instructional materials covering command-line computing, scripting, and high-performance computing workflows.
- Improved computational proficiency across the laboratory by providing hands-on technical instruction and ongoing support.

Vice President, Society of Physics Students

University of Montana | 2016–2017

Founding Member, Women in Physics

University of Montana

Selected Participant, Rutgers University Undergraduate Summer Exchange Program

Recipient, University of Montana Global Leadership Initiative Award

CONFERENCE PRESENTATIONS

Poster Presentation, Montana Space Grant Consortium Conference, 2016. Presented research on the development of a Python pipeline for Project MINERVA.

Poster Presentation, Flux Congress, 2023. Presented CABINET, a container-linking wrapper for reproducible neuroimaging workflows.

PROFESSIONAL DEVELOPMENT

Leading on All Levels, University of Minnesota (2021)

Project Management Certificate Coursework, University of Minnesota (2023)

Data Management Series, University of Minnesota (2023)